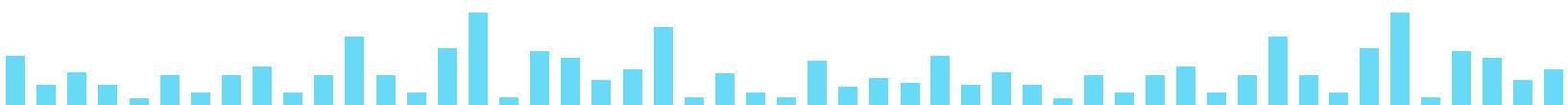
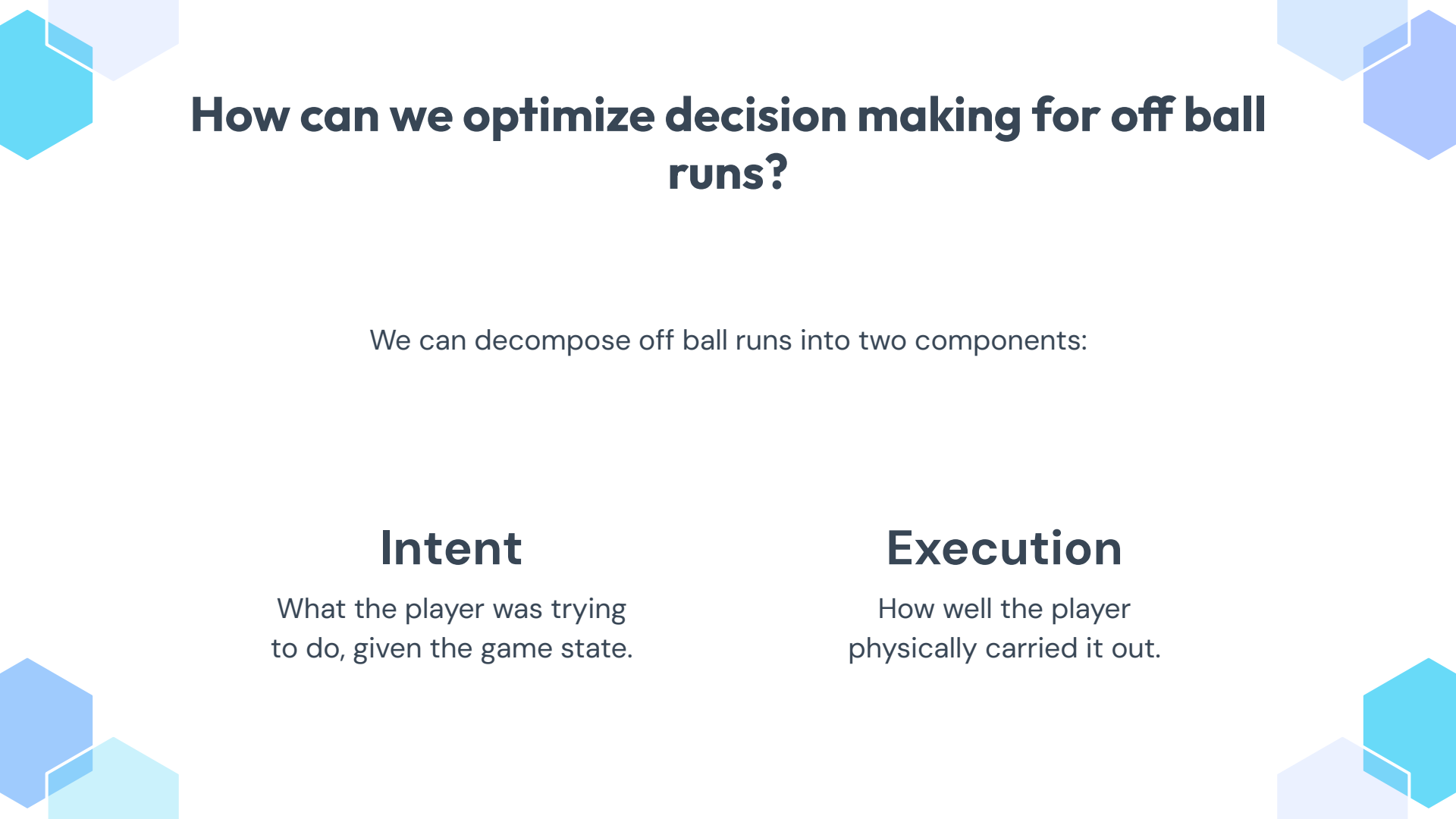




Off Ball Run Decision Making and Route Optimization

Zach Cochran





How can we optimize decision making for off ball runs?

We can decompose off ball runs into two components:

Intent

What the player was trying to do, given the game state.

Execution

How well the player physically carried it out.

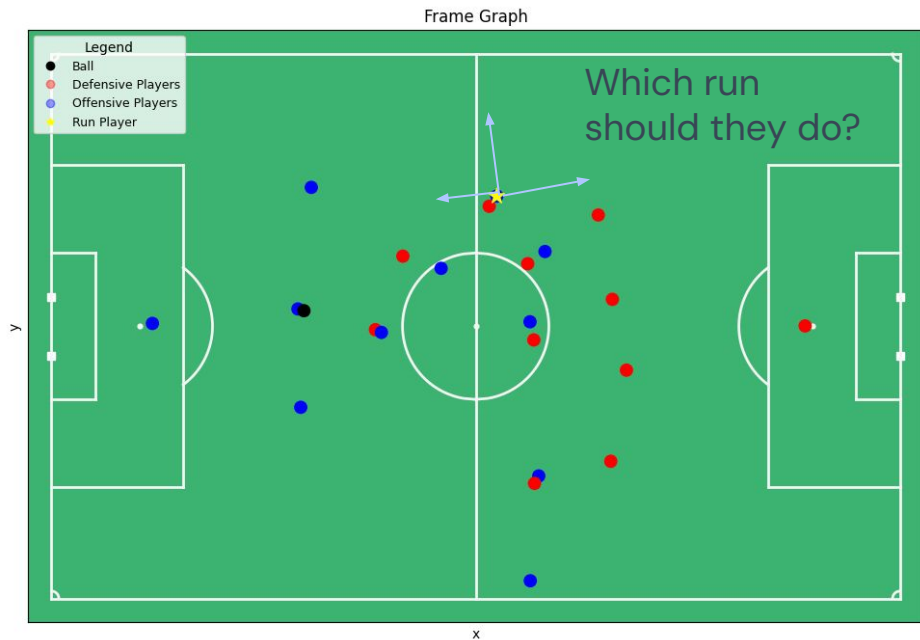
Understanding Off Ball Run Intent

Intent is the decision to make a specific type of run based on the game context.

We analyze intent by assessing:

- Viability
- Danger

Key: Maximize both viability and danger.



Understanding Off Ball Run Execution

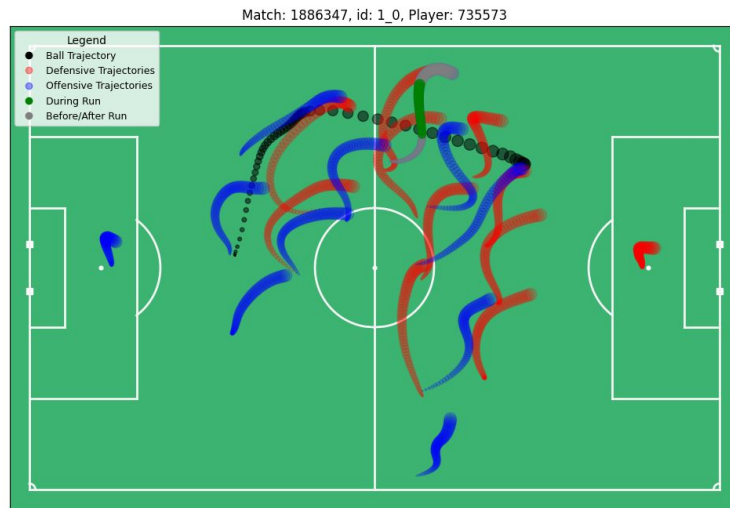
Execution refers to the trajectory taken by the runner given the intent and game situation.

Our goal is to identify trajectories that:

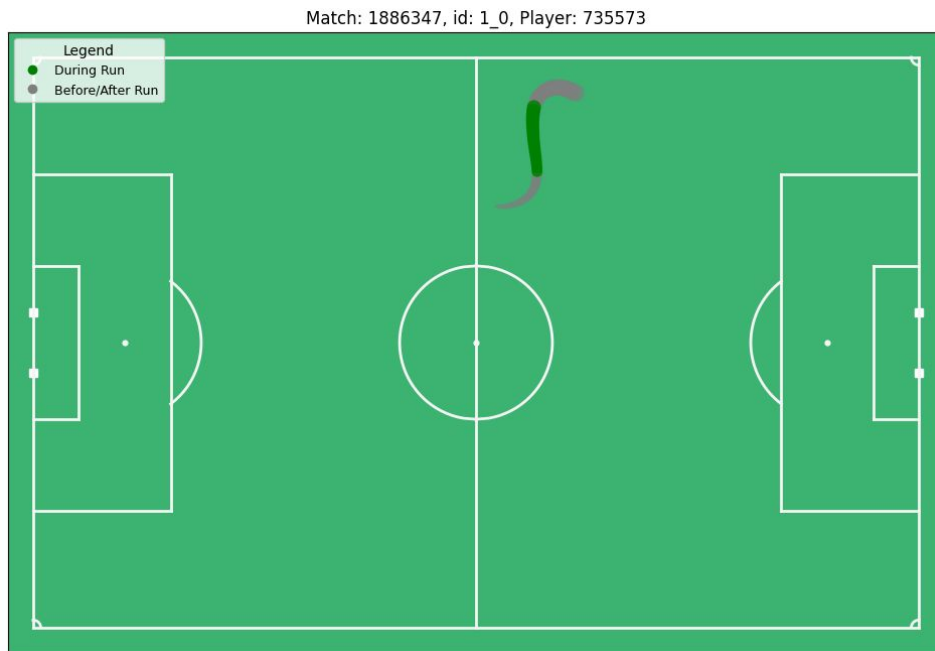
- Maximize danger
- Optimize timing
- Achieve ideal speed and acceleration profiles
- Exploit weaknesses
- Create space and opportunities

Game Situation:

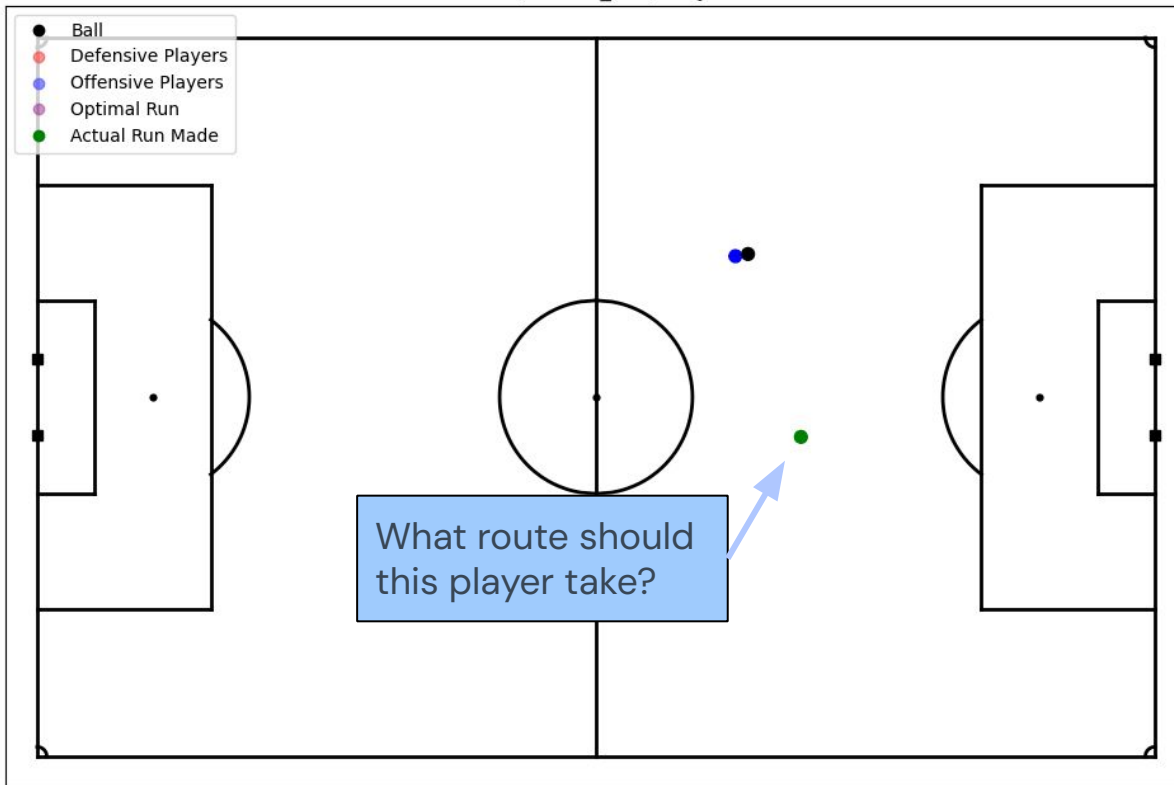
- Offensive Movement
- Defensive Movement



This project focuses on execution of off ball runs



Match: 1886347, Run: 1_202, Player: 133501.0





How?

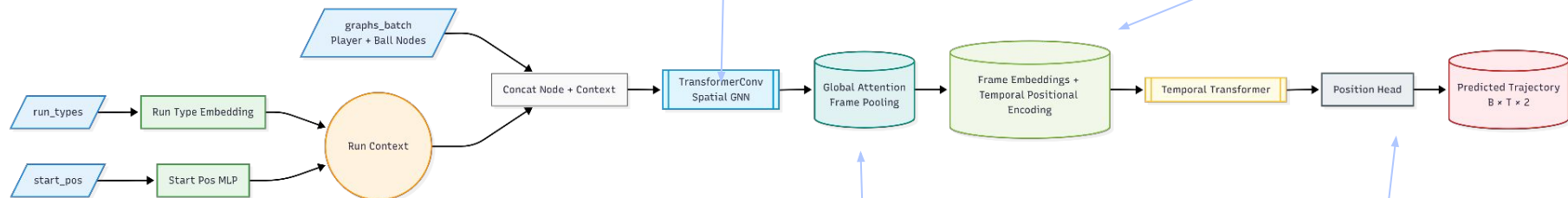
Graphs + Neural Networks



Spatial Context (Graph):

Graph Neural Network
(Player Space Summary)

Temporal Neural Network
(Space Over Time)

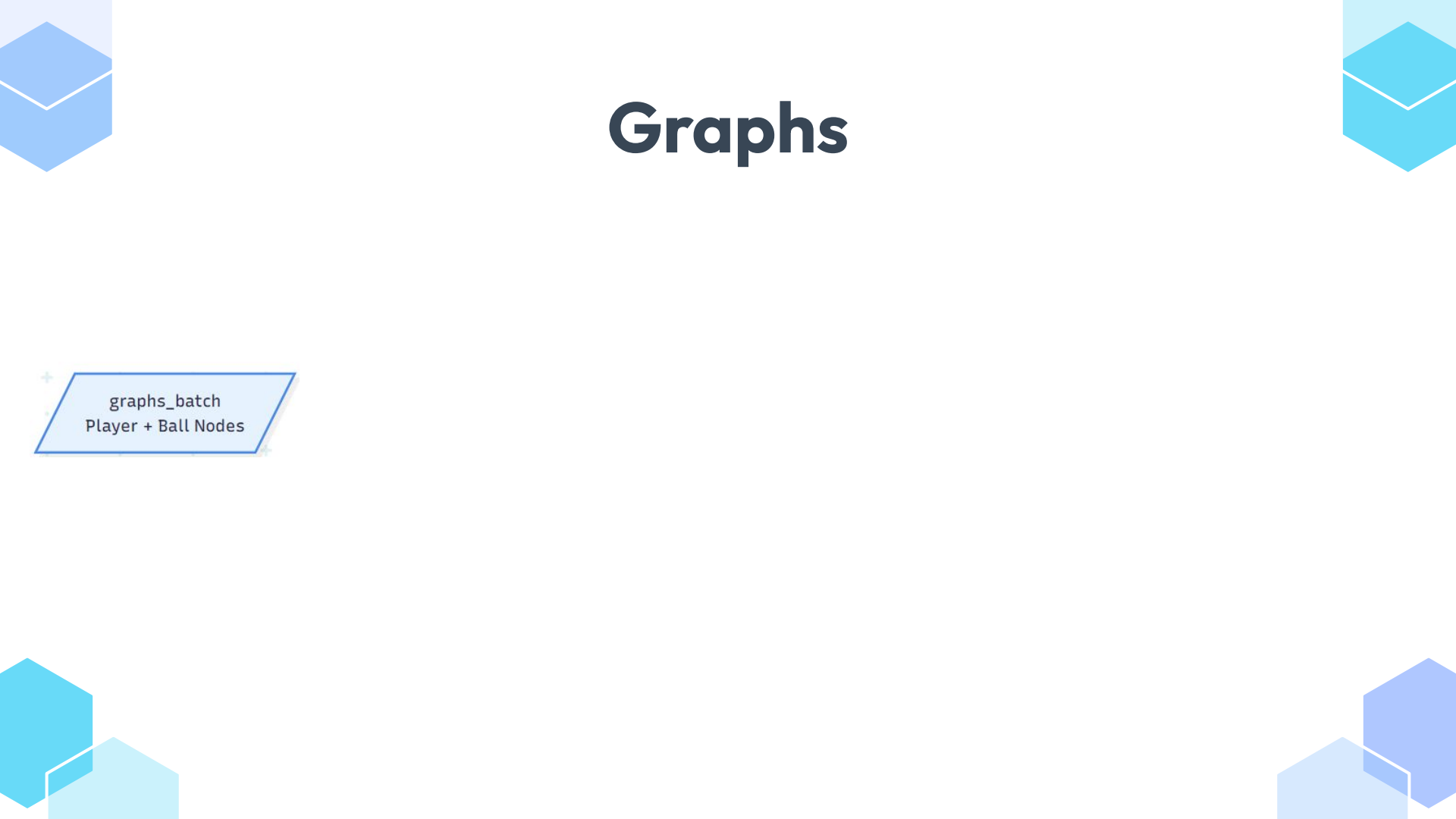


Run Context (Intent):

Global Attention Pooling
(Frame Space Summary)

Trajectory Prediction

Graphs



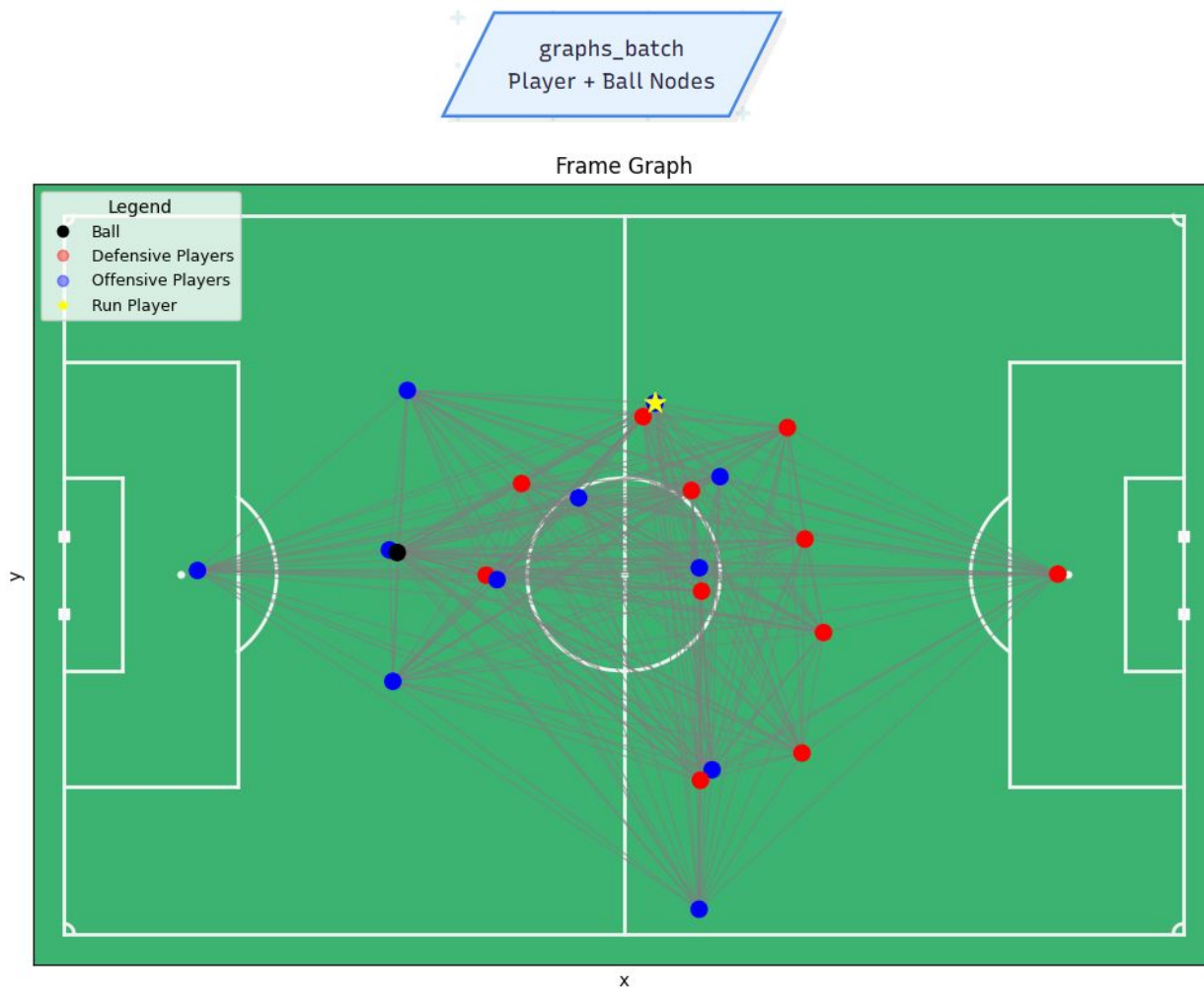
graphs_batch
Player + Ball Nodes

Graphs

Built from tracking data.

The graph encodes the spatial relationships of the players in the current frame.

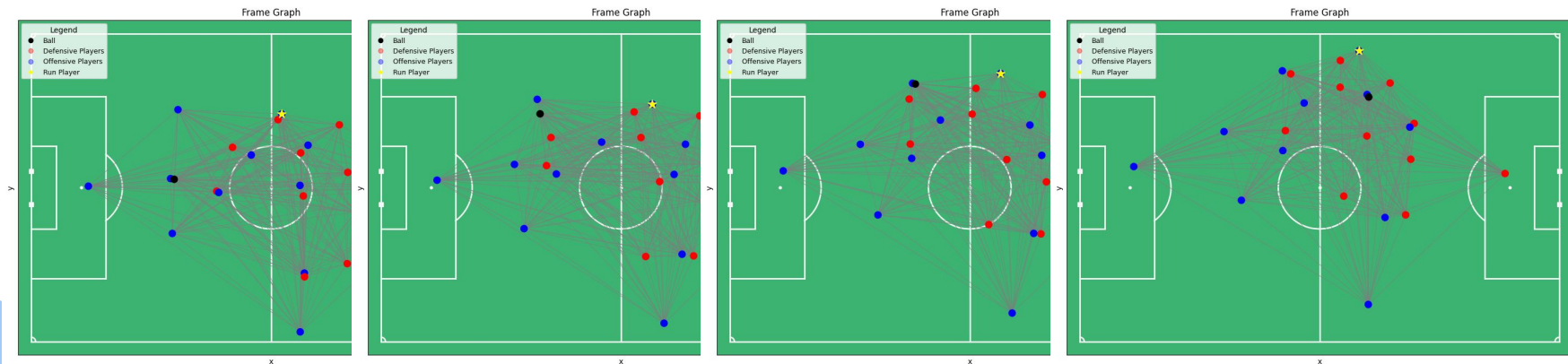
- Location
- Speed
- Relationships



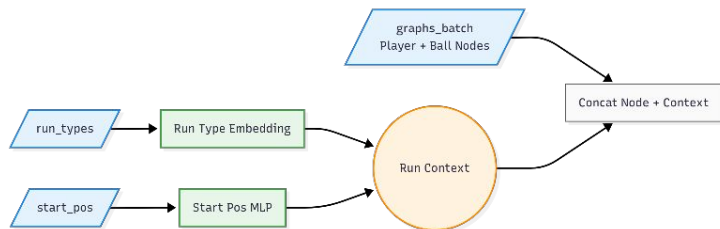
Reconstruct an Off Ball Run

graphs_batch
Player + Ball Nodes

- Sequence of graphs (Ordered list)
- Encodes defensive movement
- Encodes offensive movement
- Represents the play over time



Run Context



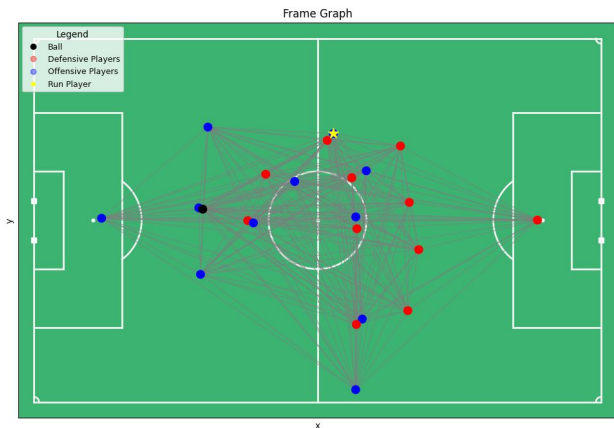
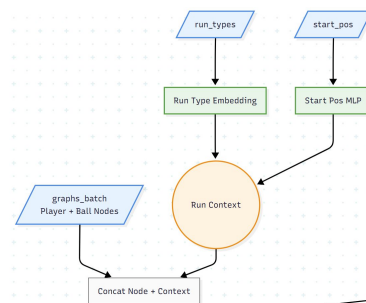
Graph + Intent

Intent:

- The run type performed from the given starting location
- Learned embedding to represent similarities in run types

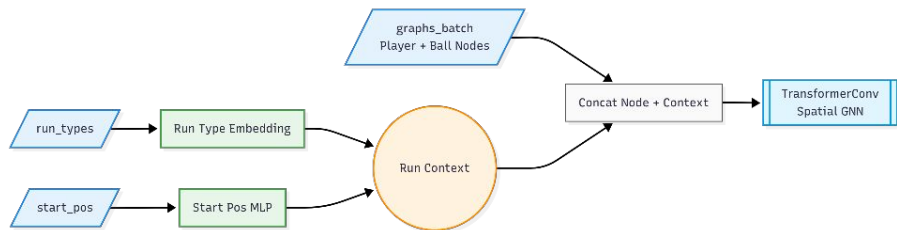
Graph + Intent:

- Combine graphs with intent as input to model.
- Learn how intent shapes run trajectories.



- Run Starting Coordinates
- Runner is trying to pull into half space

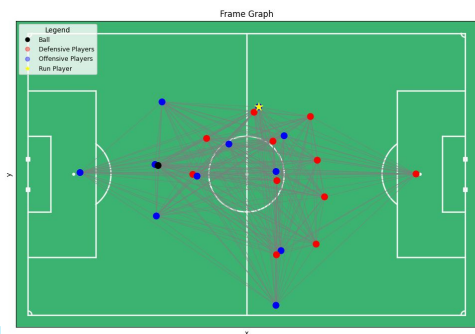
Graph Neural Network



Representing Player Spatial Context

TransformerConv
Spatial GNN

Use a graph neural network to learn how to represent the spatial context that a player is in.

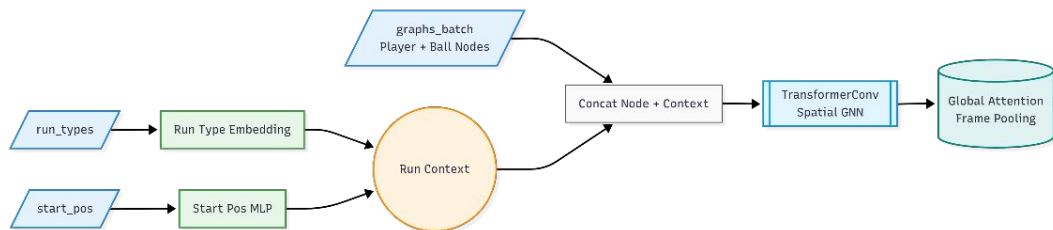


2
Transformer
Convolutions

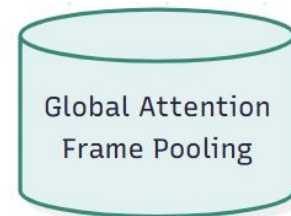
Player Features:

- Position
- Kinematics
- Local Pressure
- Teammate support
- Passing Lanes.
- Etc.

Frame Pooling



Representing Frame Wise Spatial Context



Use attention pooling to extract global features for a frame.

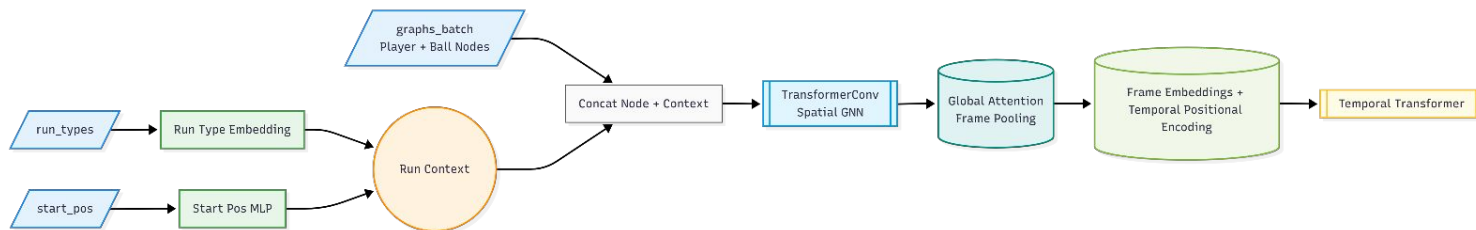
Focus attention on the spatial features/relationships of the player making the run.



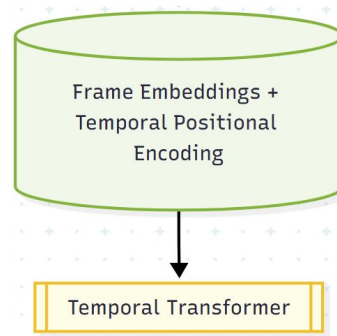
Frame Spatial Features:

- Open Space
- Counter attack / possession
- Defensive organization
- Compactness
- Overloads
- Isolations

Temporal Neural Network

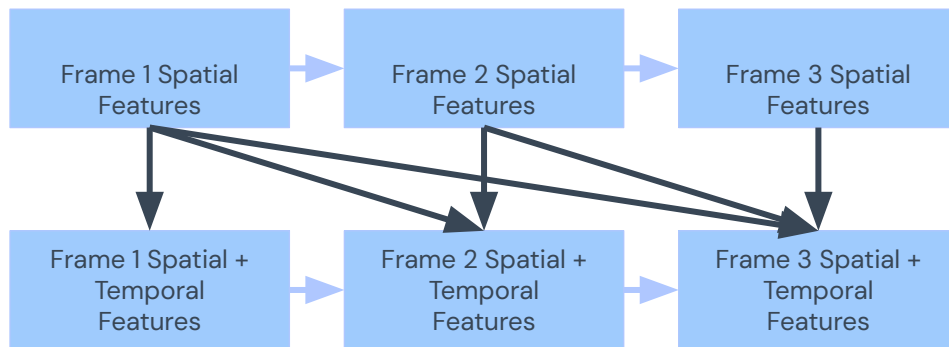


Representing Spatial Context Over Time

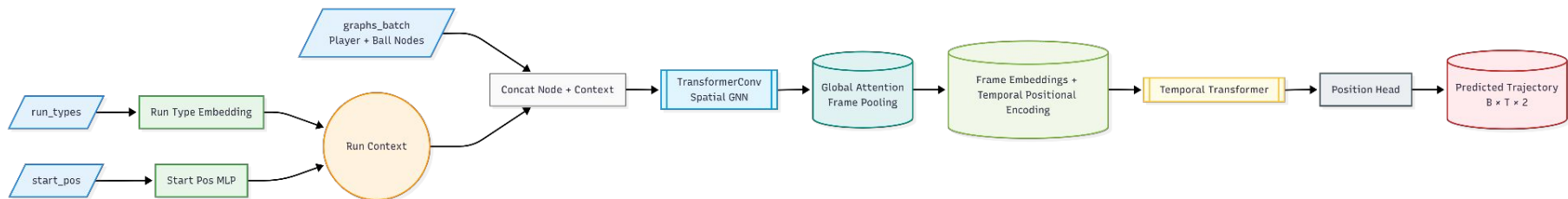


Use temporal encoders to understand how the play is unfolding over time.

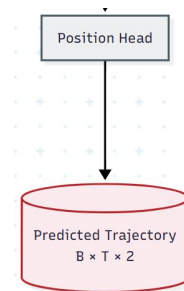
- Movement patterns
- Run timing
- Speed of play
- Space creation
- Pressure changes



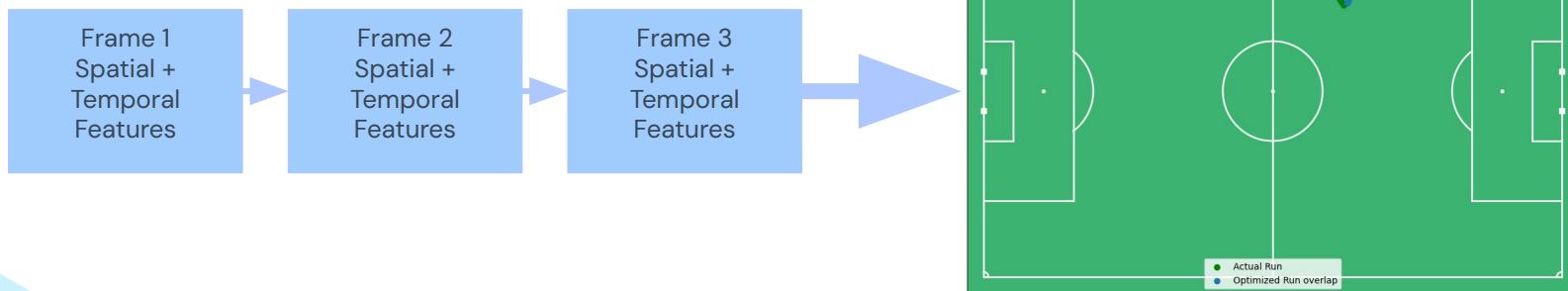
Position Head



Predicting Trajectories



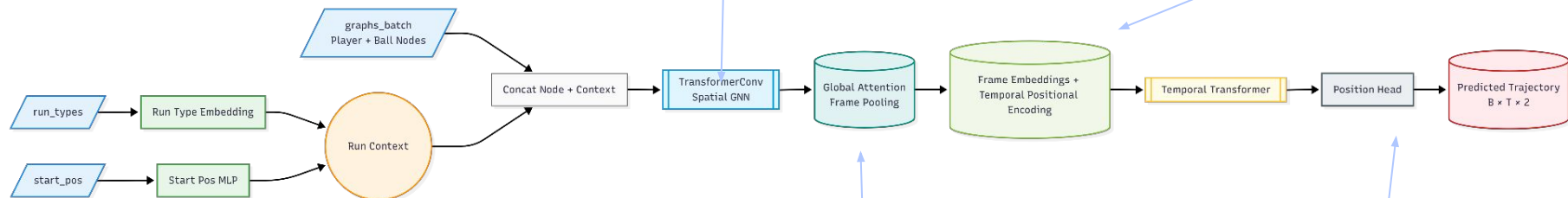
Predict run trajectories from the spatial temporal features of the run sequence.



Spatial Context (Graph):

Graph Neural Network
(Player Space Summary)

Temporal Neural Network
(Space Over Time)



Run Context (Intent):

Global Attention Pooling
(Frame Space Summary)

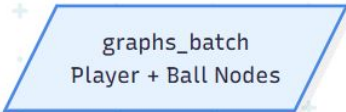
Trajectory Prediction

Model Training

Goal – To learn run movement that is:

- Realistic
- Dangerous

Key: Remove position, velocity, and acceleration of runner from graphs before input to model (We want to learn this).



graphs_batch
Player + Ball Nodes

Model Training: How?

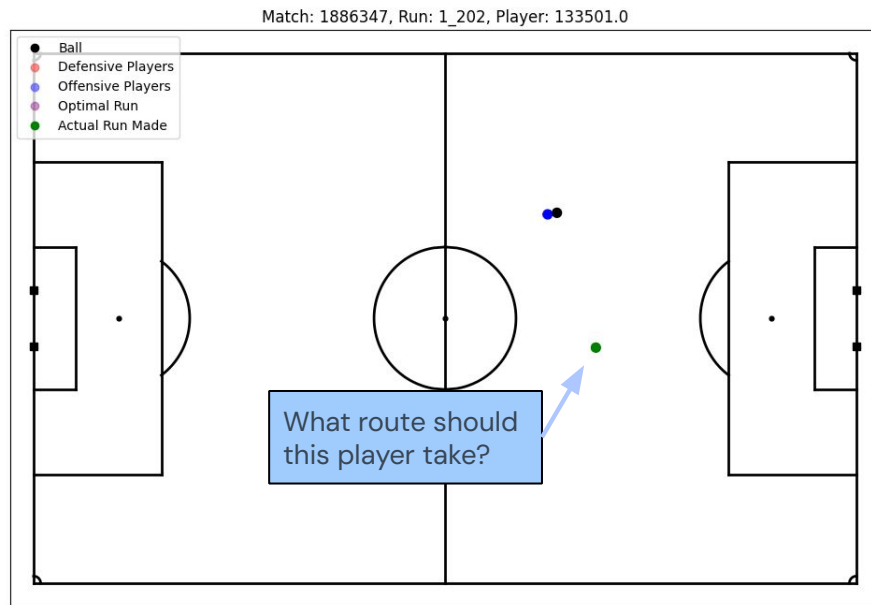
Multitask loss functions:

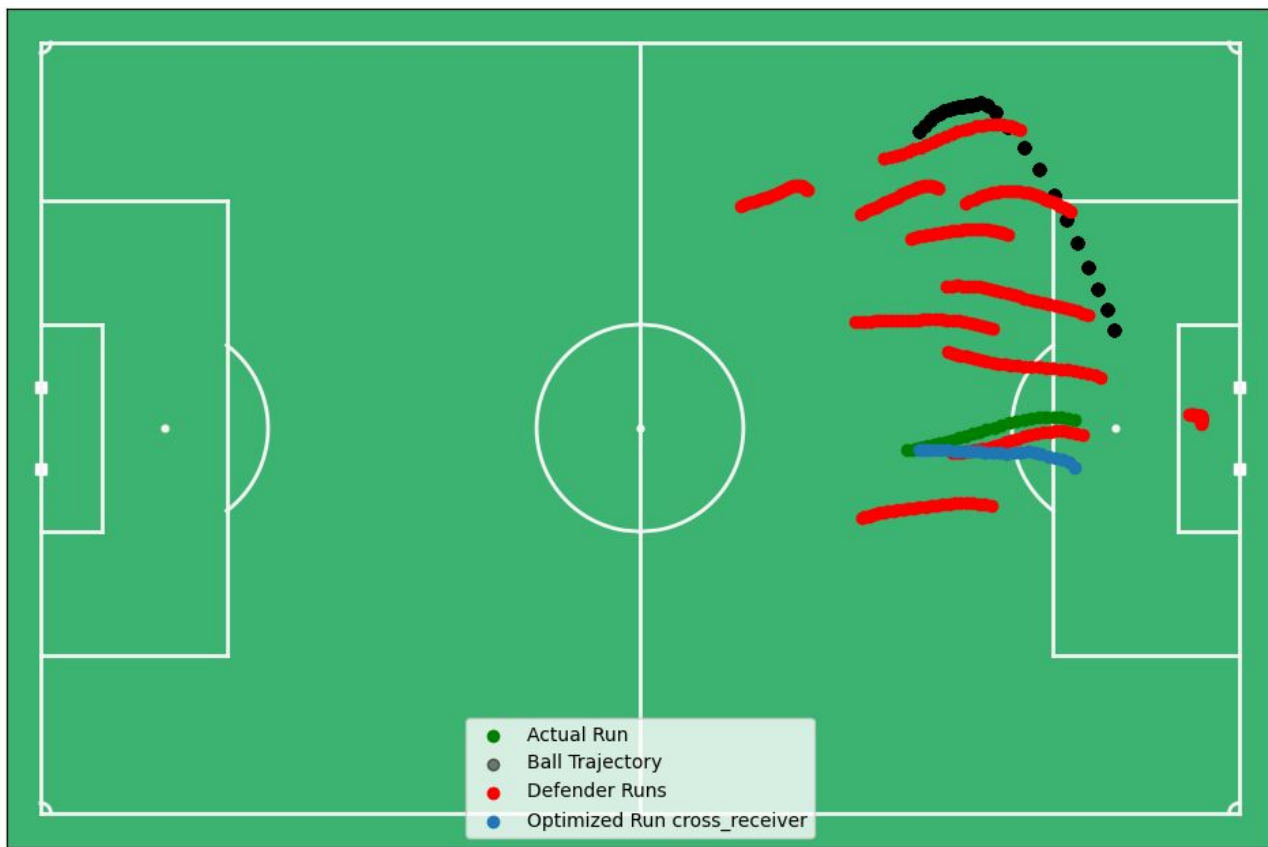
- Realistic Motion:
 - Position, Velocity and Acceleration MSE
 - Max Speed and Max Acceleration Penalty
 - Final Destination Penalty
 - Curvature Penalty – Mean Turning Angle
- Dangerous:
 - Weight loss:
 - 20x if training run lead to shot
 - 50x if training run lead to goal

Result

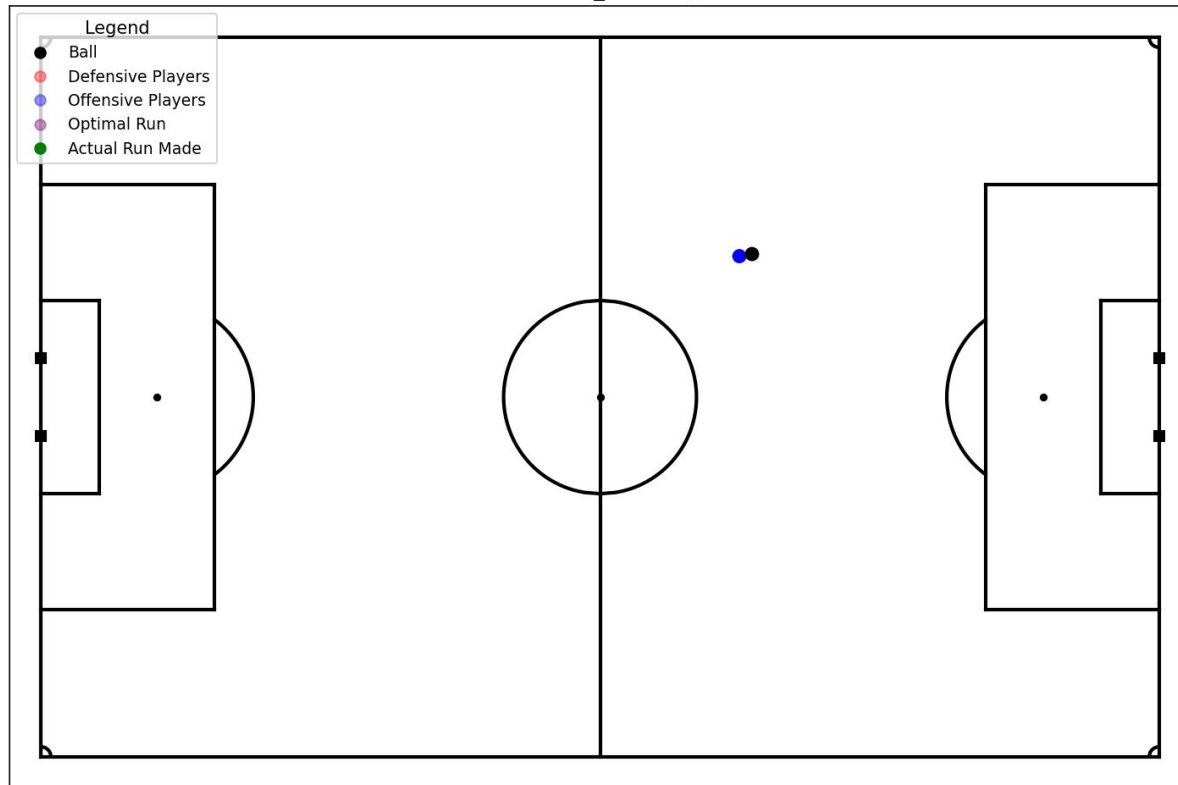
An optimized run route that considers:

- Defensive Movement
- Offensive Movement





Match: 1886347, id: 1_202, Player: 133501.0

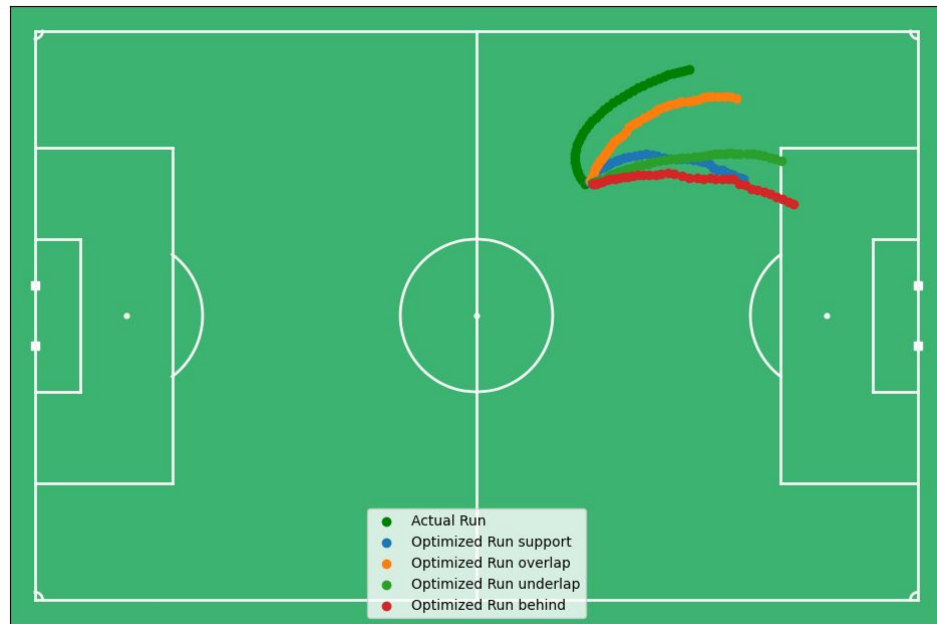


What if?

What if the runner decided to perform a different run type?

We can still find an optimal run for the new type in the same scenario.

This allows us to explore the decision making of run selection





Model Summary

We can find runs that:

- Maximize danger
- Optimize timing
- Achieve ideal speed and acceleration profiles
- Exploit weaknesses
- Create space and opportunities

We utilize:

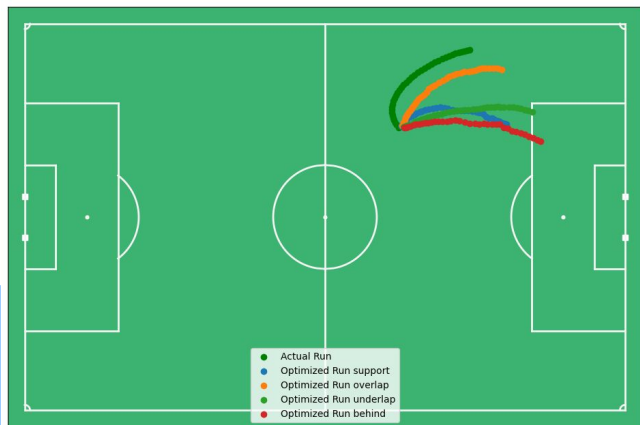
- Defensive player movements
 - Offensive player movements
 - Runner's intent
- 

Applications

- Identify where and how players should make runs
- Identify recurring patterns in missed movement
- Evaluate runs in intent vs. execution
- Counterfactual predictions (What if scenarios)
- Training and player development

Next Steps: Quantify Intent

- Find how probable each run type is given the context.
 - Ex. Run in behind is not very probable for a runner in the backline/midfield.
- Given the actual/predicted run route what is the probability of reception given the context.
 - Runs to open space for teammates vs. poorly configured runs.

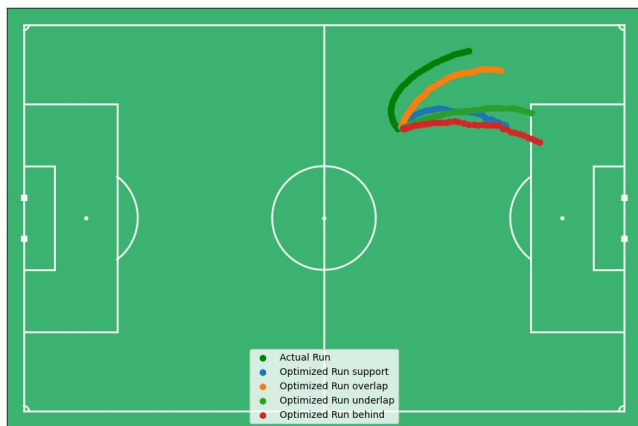


Run Type	Viability %	Reception %
Overlap	30%	70%
Underlap	20%	60%
Behind	15%	20%
Support	10%	30%

Next Steps: Quantify Execution

Find how the probability of a shot/goal occurring from the run.

- Optimized shot probability vs. actual run shot probability
- Shot probability of different types of run routes



Run Type	Lead to Shot %
Overlap	18%
Underlap	15%
Behind	25%
Support	14%
Actual Run	16%

Open Source Impact

- Objective approach to analyzing off ball runs
- Push and open research into analyzing off ball movement
- Tool for coaches and analysts:
 - Interact, visualize, and quantify decision making in off ball run scenarios
 - What if simulations
- Bridge the gap between analytics and recruiting

Changes From Original Submission

- Added intent to model input and architecture
- Removed runner position, velocity, acceleration from input graphs
- Increased weights for runs that lead to shots/goals
- Added final destination / curvature loss

Thank You!

SKILLcorner



PySport

open source